

Questions with Answer Keys

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Q1 - Single Correct

The bottom of legs of a three legged table are the vertices of an isosceles triangle with sides 5,5 and 6. The legs are to be braced at the bottom by three wires in the shape of a Y. The minimum length of the wire needed for this purpose is

(1) $4 + 3\sqrt{3}$

(2) 10

(3) $3 + 4\sqrt{3}$

(4) $1 + 6\sqrt{2}$

Q2 - Single Correct

A horse runs along a circle with a speed of 20 km/h. A lantern is at the centre of the circle. A fence is along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse moves along the fence at the moment when it covers $1/8$ of the circle in km/h, is

(1) 20

(2) 40

(3) 30

(4) 60

Q3 - Single Correct

Let $f(x)$ and $g(x)$ be two continuous functions defined from $R \rightarrow R$, such that $f(x_1) > f(x_2)$ and $g(x_1) < g(x_2), \forall x_1, x_2$, then solution set of $f[g(\alpha^2 - 2\alpha)] > f[g(3\alpha - 4)]$ is

(1) R

(2) ϕ

(3) (1, 4)

(4) $R - [1, 4]$

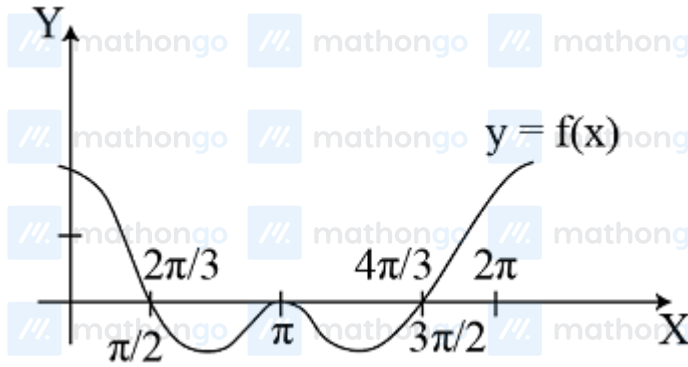
Q4 - Single Correct

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The graph of $y = f(x)$ is shown. Let $F(x)$ be an anti-derivative of $f(x)$. Then, $F(x)$ has



(1) points of inflexion at $x = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$ and 2π , a local maximum at $x = \pi/2$ and a local minimum at

$x = 3\pi/2$

(2) points of inflexion at $x = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$ and 2π , a local minimum at $x = \pi/2$ and a local maximum at

$x = \frac{3\pi}{2}$

(3) point of inflexion at $x = \pi$, a local maximum at $x = \pi/2$ and a local minimum at $x = 3\pi/2$

(4) points of inflexion at $x = \pi$, a local minimum at $x = \pi/2$ and a local maximum at $x = 3\pi/2$

Q5 - Single Correct

The number of critical points of the function, $f(x) = \frac{2}{3}\sqrt{x^3} - \frac{x}{2} + \int_1^x \left(\frac{1}{2} + \frac{1}{2}\cos 2t - \sqrt{t} \right) dt$ which lie in the interval $[-2\pi, 2\pi]$, is

(1) 2

(2) 4

(3) 6

(4) 8

Q6 - Single Correct

If $f'(x) = |x| - \{x\}$, where $\{ \cdot \}$ denotes fractional part of x , then $f(x)$ is decreasing in

(1) $(-1/2, 0)$

(2) $(1/2, 2)$

(3) $(-1/2, 2)$

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(4) $(1/2, \infty)$

Q7 - Single Correct

If $M(x_0, y_0)$ is the point on the curve $3x^2 - 4y^2 = 72$, which is nearest to the line $3x + 2y + 1 = 0$, then the value of $(x_0 + y_0)$ is equal to

(1) 3

(2) -3

(3) 9

(4) -9

Q8 - Single Correct

If $f'(x) > 0$, $f''(x) > 0, \forall x \in R$, then for any two real numbers x_1 and x_2 ($x_1 \neq x_2$)

(1) $f\left(\frac{x_1+x_2}{2}\right) > \frac{f(x_1)+f(x_2)}{2}$

(2) $f\left(\frac{x_1+x_2}{2}\right) = \frac{f(x_1)+f(x_2)}{2}$

(3) $f\left(\frac{x_1+x_2}{2}\right) < \frac{f(x_1)+f(x_2)}{2}$

(4) None of these

Q9 - Single Correct

Let $f'(\sin x) < 0$, $f''(\sin x) > 0, \forall x \in \left(0, \frac{\pi}{2}\right)$ and $g(x) = f(\sin x) + f(\cos x)$, then the interval in which $g(x)$ is increasing, is

(1) $(\pi/4, \pi)$ (2) $(\pi, 3\pi/2)$ (3) $(\pi/4, \pi/2)$ (4) $(0, \pi/4)$

Q10 - Single Correct

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If $f(x) = \begin{cases} x-1, & x \geq 1 \\ [a+1] - x^3 + 1, & x < 1 \end{cases}$ has minimum at $x = 1$, then (where $[\cdot]$ denotes the greatest integer function)

(1) $a \geq -1$

(2) $a \leq -1$

(3) $-1 \leq a \leq 1$

(4) $a \neq 0$

Q11 - Single Correct

$f(x)$ is twice differentiable function such that $f(x) + f''(x) = -x|\sin x|f'(x)$, where $x \geq 0$. Given that $f(0) = -3, f'(0) = 4$ the maximum value of $f(x)$ is

(1) 4

(2) 3

(3) 5

(4) Does not exist

Q12 - Single Correct

If $a, b \in \mathbb{R}$ distinct numbers satisfying $|a-1| + |b-1| = |a| + |b| = |a+1| + |b+1|$, then the minimum value of $|a-b|$ is

(1) 3

(2) 0

(3) 1

(4) 2

Q13 - Single Correct

If $f(x) = f(x-1)|x^2 - 2007x + 2010| + \sin[(x-1)]\log_e x$ has

(1) local minimum at $x = 1$

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(2) local maximum at $x = 1$ (3) no local extremum at $x = 1$

(4) None of these

Q14 - Single Correct

The function $f(x) = |px - q| + r|x|$, $x \in (-\infty, \infty)$, where $p > 0, q > 0, r > 0$ assumes its minimum value only on one point of

(1) $p \neq q$ (2) $r \neq q$ (3) $r \neq p$ (4) $p = q = r$

Q15 - Multiple Correct

If $f : R \rightarrow R$ and $g : R \rightarrow R$ are two functions such that $f(x) + f''(x) = -xg(x) \cdot f'(x)$ and $g(x) > 0$,

$\forall x \in R$, then the functions $f^2(x) + [f'(x)]^2$ has

(1) a maximum at $x = 0$ (2) a minimum at $x = 0$ (3) a point of inflexion at $x = 0$

(4) None of these

Q16 - Multiple Correct

The function $f(x) = \begin{cases} x + 2, & \text{if } x < -1 \\ x^2, & \text{if } -1 \leq x < 1 \\ (x - 2)^2, & \text{if } x \geq 1 \end{cases}$

(1) is continuous for all $x \in R$ (2) is continuous but not differentiable for all $x \in R$ (3) is such that $f'(x)$ change its sign exactly twice

(4) has two local maxima and two local minima

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Q17 - Multiple Correct

Which of the following inequalities always hold good in $(0, 1)$?

(1) $x > \tan^{-1} x$

(2) $\cos x < 1 - \frac{x^2}{2}$

(3) $1 + x \log(x + \sqrt{1 + x^2}) > \sqrt{1 + x^2}$

(4) $x^2 - \frac{x^2}{2} < \log(1 + x)$

Q18 - Multiple Correct

If $f(x)$ is continuous and differentiable, $\forall x \in R$ and $f'(c) = 0$ for exactly two real values of c , then the number of real and distinct values of d for which $f(d) = 0$ can be

(1) 1

(2) 2

(3) 3

(4) 4

Q19 - Multiple Correct

If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$ (a is a finite quantity), (where $[\cdot]$ denotes the greatest integer function), then

(1) $\lim_{x \rightarrow a} f(x)$ is an integer

(2) $\lim_{x \rightarrow a} f(x)$ needed not be an integer

(3) $f(x)$ has a local minima at $x = a$

(4) $f(x)$ has a local maxima at $x = a$

Q20 - Multiple Correct

Which of the following functions fail to satisfy the condition of Rolle's theorem on the interval $[-1, 1]$?

(1) $f(x) = |x|[x]$

(2) $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

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(3) $f(x) = \{x\} + \{-x\}$

(4) $f(x) = |x| - \sin|x|$

Q21 - Multiple Correct

Let $f(x) = 1 + x \log(x + \sqrt{x^2 + 1}) - \sqrt{1 + x^2}$ and $h(x) = f(x) - f^2(x) + f^3(x)$. Then

(1) $h(x)$ is increasing in $(0, \infty)$ (2) $h(x)$ is decreasing in $(-\infty, 0)$ (3) $h(x)$ is increasing on R (4) $h(x)$ is decreasing on R

Q22 - Multiple Correct

Let $f(x) = a(x - x_1)(x - x_2)$, where a, x_1 and x_2 are real numbers such that $a \neq 0$ and $x_1 + x_2 = 0$. Then, which of the following statements is/are always correct?

(1) $f(x)$ is increasing in $(-\infty, 0)$ and decreasing in $(0, \infty)$ (2) $f(x)$ is decreasing in $(-\infty, 0)$ and increasing in $(0, \infty)$ (3) $f(x)$ is non-monotonic on R (4) $f(x)$ has an extremum point

Q23 - Multiple Correct

A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then, the lengths of sides of the rectangular sheet are

(1) 24

(2) 32

(3) 45

(4) 60

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Q24 - Multiple Correct

$f(x)$ is cubic polynomial which has local maximum at $x = -1$. If $f(2) = 18$, $f(1) = -1$ and $f'(x)$ has local minima at $x = 0$, then

(1) the distance between $(-1, 2)$ and $(a, f(a))$, where $x = a$, the point of local minima is $2\sqrt{5}$

(2) $f(x)$ is increasing for $x \in [1, 2\sqrt{5}]$

(3) $f(x)$ has local minima at $x = 1$

(4) has value of $f(0) = 5$

Q25 - Multiple Correct

For the function $f(x) = x \cos(1/x)$, $x \geq 1$

(1) for atleast one x in the interval $[1, \infty)$, $f(x+2) - f(x) < 2$

(2) $\lim_{x \rightarrow \infty} f'(x) = 1$

(3) for all x in the interval $[1, \infty)$, $f(x+2) - f(x) > 2$

(4) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$

Q26 - Multiple Correct

If $f(0) = f(1) = f(2) = 0$ and function $f(x)$ is twice differentiable in $(0, 2)$ and continuous in $[0, 2]$. Then, which of the following is/are true?

(1) $f''(c) = 0, \forall c \in (0, 2)$

(2) $f'(c) = 0$, for atleast two $c \in (0, 2)$

(3) $f'(c) = 0$, for exactly one $c \in (0, 2)$

(4) $f''(c) = 0$, for atleast one $c \in (0, 2)$

Q27 - Multiple Correct

Water is poured into an inverted conical vessel of which the radius of the base is 2 m and height 4 m at the rate of 77 L/min. The rate at which the water level is rising at the instant when the depth is 70 cm, is

(1) 10 cm/min

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(2) 20 cm/min

(3) 40 cm/min

(4) None of these

Q28 - Multiple Correct

The abscissa of the point on the curve $\sqrt{xy} = a + x$, ($a > 0$) the tangent at which cuts off equal intercepts from the coordinates axes is

(1) $a/\sqrt{2}$ (2) $-a/\sqrt{2}$ (3) $a\sqrt{2}$ (4) $-a\sqrt{2}$

Q29 - Paragraph 1

Passage I (For Question 29, 30)

Let $f(x)$ be a real valued continuous function on R defined as $f(x) = x^2 e^{-|x|}$.

The value of k for which the curve $y = kx^2$ ($k > 0$) intersect the curve $y = e^{|x|}$ at exactly two points, is

(1) e^2 (2) $e^2/2$ (3) $e^2/4$ (4) $e^2/8$

Q30 - Paragraph 1

The area bounded by ordinates at points of local maximum, the curve $y = f(x)$ and X -axis, is

(1) $\frac{4(e^2-5)}{e^2}$ (2) $\frac{e^2+5}{e}$ (3) $\frac{2(e^2+5)}{e^2}$

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(4) 4

Q31 - Paragraph 2

Passage II (For Question 31, 32, 33)

Consider the functions $f(x) = \frac{x}{2^x}$ and $g(x) = \max\{f(t) : x \leq t \leq x+1\}$.

Which of the following statement is incorrect?

- (1) $f(x)$ is increasing in $\left(-\infty, \frac{1}{\log_e 2}\right)$ and decreasing in $\left(\frac{1}{\log_e 2}, \infty\right)$
- (2) Local maximum value of $f(x)$ exists
- (3) Local minimum value of $f(x)$ does not exist
- (4) $f(x)$ changes its concavity at 2 points

Q32 - Paragraph 2

If $f(x) = k$ has 2 distinct real roots, then the range of k is

- (1) $(0, 1/e)$
- (2) $\left(0, \frac{1}{e \log_e 2}\right)$
- (3) $\left(\frac{1}{e \log_e 2}, \infty\right)$
- (4) $(-\infty, 0)$

Q33 - Paragraph 2

The value of function $g(x)$ is

- (1) $(x+1)2^{-x+1}$
- (2) $(x-1)2^{-x-1}$
- (3) $x \cdot 2^{-x}$
- (4) $(x+1)2^{-(x+1)}$

Q34 - Paragraph 3

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Passage III (For Question 34, 35)

Suppose $f(x)$ is a real valued polynomial function of degree 6 satisfying the following conditions.

i. f has minimum value at $x = 0$ and 2.

ii. f has maximum value at $x = 1$.

iii. For all x , $\lim_{x \rightarrow 0} \frac{1}{x} \log \begin{vmatrix} \frac{f(x)}{x} & 1 & 0 \\ 0 & \frac{1}{x} & 1 \\ 1 & 0 & \frac{1}{x} \end{vmatrix} = 2$.

The number of solution of the equation $8f(x) - 1 = 0$ is

(1) 1

(2) 2

(3) 3

(4) 4

Q35 - Paragraph 3

The range of $f(x)$ is

(1) $\left[-\frac{32}{15}, \infty\right)$

(2) $\left[-\frac{4}{15}, \infty\right)$

(3) $\left(-\infty, \frac{2}{15}\right]$

(4) None of these

Q36 - Integer Type

Suppose that f is differentiable for all x , such that $f'(x) \leq 2$ for all x . If $f(1) = 2$ and $f(4) = 8$, then $f(2)$

has the value equal to

Q37 - Integer Type

If $f(x)$ and $g(x)$ are two differentiable functions in $[0, 2]$ such that $f''(x) - g''(x) = 0$,

$f'(1) = 2, g'(1) = 4, f(2) = 3, g(2) = 9$, then $|f(x) - g(x)|$ at $3/2$ is equal to

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Q38 - Integer Type

If $x + 4y = 14$ is normal to the curve $y^2 = \alpha x^3 - \beta$ at $(2, 3)$, then the value of $(\alpha + \beta)$ is

Q39 - Integer Type

The angle of intersection of the curves $y = [|\sin x| + |\cos x|]$ and $x^2 + y^2 = 5$, where $[\cdot]$ denotes the greatest integer function is $\tan^{-1}(k)$. Then, k is equal to

Q40 - Integer Type

A man 1.6 m high walks at the rate of 30 m/min away from a lamp which is 4 m above ground. If k is the rate at which the man's shadow is lengthening, then the value of $\frac{k}{4}$ is

Q41 - Integer Type

Let P be the point on the curve $4x^2 + \alpha^2 y^2 = 4\alpha^2$, $0 < \alpha^2 < 8$, whose distance from $Q(0, -2)$ is greatest. If R is the reflection of P in the X -axis, then the least distance of R from the line $3x - 4y + 7 = 0$ is

Q42 - Integer Type

The minimum value of the function $f(x) = \frac{\tan(x+\pi/6)}{\tan x}$ is

Q43 - Integer Type

The minimum distance between the curves $f(x) = e^x$ and $g(x) = \log_e x$ is $\sqrt{k}$, then k is

Q44 - Integer Type

The lower corner of a leaf in a book is folded over, so as to just reach the inner edge of the page. The fraction of width folded over, if the area of the folded part is minimum is p/q , where $p, q \in N$, then the least value of $(p + q)$ is

Q45 - Integer Type

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A parabola of the form $y = ax^2 + bx + c$ with $a > 0$ intersects the graph of $f(x) = \frac{1}{x^2 - 4}$. Then, the minimum number of points of intersection is

Q46 - Integer Type

If the largest positive value of the function $f(x) = \sqrt{8x - x^2} - \sqrt{14x - x^2 - 48}$ is $\sqrt{k}$, where $k \in N$, then the number of positive divisors of k is

Questions with Answer Keys

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Answer Key

Q1 (1)	Q2 (2)	Q3 (3)	Q4 (3)
Q5 (2)	Q6 (1)	Q7 (2)	Q8 (3)
Q9 (3)	Q10 (1)	Q11 (3)	Q12 (4)
Q13 (3)	Q14 (3)	Q15 (1)	Q16 (1, 2, 4)
Q17 (1, 3, 4)	Q18 (1, 2, 3)	Q19 (1, 3)	Q20 (1, 2, 3)
Q21 (1, 2)	Q22 (3, 4)	Q23 (1, 3)	Q24 (2, 3)
Q25 (2, 3, 4)	Q26 (2, 4)	Q27 (2)	Q28 (1, 2)
Q29 (3)	Q30 (1)	Q31 (4)	Q32 (2)
Q33 (4)	Q34 (4)	Q35 (1)	Q36 (4)
Q37 (5)	Q38 (9)	Q39 (2)	Q40 (5)
Q41 (3)	Q42 (3)	Q43 (2)	Q44 (5)
Q45 (2)	Q46 (6)		